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Viromic analysis of cerebrospinal fluid (CSF) samples identified an unclassified RNA virus exhibiting an atypical genomic architecture

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Abstract

The central nervous system (CNS) is highly susceptible to infections that can lead to severe neurological morbidity and mortality. Despite advances in diagnostic technologies, a significant proportion of encephalitis and meningitis cases remain etiologically undiagnosed, underscoring the need for novel pathogen discovery approaches. Here, we report the identification of a novel unclassified RNA virus, provisionally named hucaurvirus ("human CSF-associated unclassified RNA virus"), in the cerebrospinal fluid (CSF) of a pediatric patient with unexplained meningoencephalitis, using metagenomic next-generation sequencing (mNGS). The hucaurvirus exhibits a monopartite, positive-sense RNA genome of 6.5 kb which contains two overlapping open reading frames (ORFs), which was confirmed by conventional PCR amplification and Sanger sequencing. The large ORF encodes a polyprotein containing a capsid domain, while the small ORF overlaps within the large ORF and encodes the RNA-dependent RNA polymerase (RdRp) protein. BLASTp search based on the amino acid sequence of RdRp showed that hucaurvirus had the highest sequence identity and query coverage of 39.2% and 90%, respectively, to a noda-like virus (GenBank no. MT138110). Phylogenetic analysis showed that hucaurvirus was closely related to four viruses annotated as noda-like viruses, forming a cluster that is adjacent to the cluster of the *Carmotetraviridae* family and appears to form a new viral family within the order *Tolivirales*. PCR screening confirmed the presence of hucaurvirus in one of ten CSF samples, validating its detection. Our findings emphasize the utility of mNGS in uncovering novel pathogens and expanding our understanding of viral diversity in CNS infections.

Keywords Hucaurvirus, Unclassified RNA virus, Genome, Phylogenetic analysis

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Introduction

The central nervous system (CNS) represents one of the most critical and protected biological compartments in humans, yet it remains vulnerable to infections that can lead to severe neurological morbidity and mortality. Despite advances in diagnostic technologies, a significant proportion of encephalitis and meningitis cases—up to 60% in some cohorts—remain etiologically undiagnosed, highlighting a critical gap in our understanding of neurotropic pathogens [1, 2]. Traditional pathogen detection methods, such as culture-based techniques and targeted PCR assays, are inherently limited by their reliance on prior knowledge of suspected agents, leaving novel or divergent viruses undetected. In recent years, metagenomic next-generation sequencing (mNGS) has emerged as a transformative tool for unbiased pathogen discovery, enabling the identification of novel viral sequences directly from clinical samples without prior assumptions about their identity [3]. This approach has proven particularly valuable in studying cerebrospinal fluid (CSF), a diagnostically rich but low-biomass sample type, where conventional methods often fail to detect fastidious or unexpected pathogens [3, 4].

RNA viruses encompass a vast and diverse group of pathogens that infect a wide range of hosts, spanning from prokaryotes to eukaryotes, including fungi, plants, and animals [5]. In recent years, with the advent of advanced sequencing technologies such as mNGS, a growing number of unclassified RNA viruses have been discovered, expanding our understanding of the viral world's complexity [6–8]. Here, we report the discovery and characterization of a highly divergent RNA virus from the CSF sample of a patient with unexplained meningoencephalitis, provisionally named hucaurvirus, which is derived from the abbreviation of “human CSF-associated unclassified RNA virus”. Through viral metagenomics and phylogenetic analysis, we demonstrate that hucaurvirus shares limited sequence homology with some unclassified RNA viruses annotated as noda-like virus and some classified members within the *Carmotetraviridae* family. Notably, hucaurvirus exhibits a genome organization distinct from other unclassified RNA viruses and members in *Carmotetraviridae* and possesses a large ORF and a small ORF overlapping within the large ORF, which encodes polyprotein and RdRp, respectively. Phylogenetic reconstruction places hucaurvirus on a lineage containing several noda-like viruses and is adjacent to the cluster of the *Carmotetraviridae*, seeming to form a new viral family within the order *Tolivirales*.

Materials and methods

Samples treatment

For viral metagenomic analysis, 10 CSF specimens were collected from pediatric patients (aged 3 to 6 years old) with encephalitis of unknown etiology, who were admitted to Taizhou People's Hospital in China in May to Jun in 2023 (detailed information in Table S1). The CSF samples were pooled into one specimen pool and thoroughly mixed, resuspended in 1 mL of DPBS, and subjected to three rapid freeze-thaw cycles on dry ice to release the viral particles that may be hidden inside intact cells in the sample. The supernatants were collected after centrifugation (10 min, 15,000×g) and filtered through a 0.45 µm filter (Millipore) to remove eukaryotic and bacterial cell-sized particles [9]. A 200 µL aliquot of the supernatant from the pool was treated with a mixture of DNases [Turbo DNase (Ambion), Baseline-ZERO (Epicentre) and benzonase (Novagen)] and RNase (Fermentas) at 37 °C for 60 min to reduce the concentration of free (non-viral encapsidated) nucleic acids [10]. The remaining total nucleic acids were isolated using the QIAamp MinElute Virus Spin Kit (Qiagen) according to the manufacturer's protocol. This study obtained approval from the Ethics Committee of Jiangsu University (Approval No. JSDX1007002).

Library construction and sequencing

These extracted nucleic acid samples containing DNA and RNA viral sequences were used for reverse transcription reactions with the SuperScript III reverse transcriptase (Invitrogen) and 100 pmol of a random hexamer primer, followed by a single round of DNA synthesis using Klenow fragment polymerase (New England BioLabs). Briefly, total nucleic acid sample was mixed with 50 pmol of random hexamer primer, denatured at 75 °C for 5 min, and chilled on ice. A reaction mix containing 4 µL of 5× first-strand buffer (Invitrogen), 1 µL of 100 mM dithiothreitol (DTT), 1 µL solution containing each deoxynucleoside triphosphate (dNTP) at 10 mM, and 200 units of SuperScript III reverse transcriptase (Invitrogen) was added. The total of 20 µL reaction mixture was incubated at 25 °C for 10 min and then at 42 °C for 60 min. After a denaturation step at 92 °C for 5 min and chilling on ice (to reanneal random hexamer primer to cDNA), 5 units (1 µL) of Klenow fragment polymerase was added and incubated at 37 °C for 1 h, followed by an enzyme inactivation step at 75 °C for 10 min. The library was constructed using the Nextera XT DNA Sample Preparation Kit (Illumina) and sequenced on the Illumina NovaSeq platform with a 150 bp paired-end sequencing strategy.

Bioinformatic analyses

The 150 bp paired-end reads generated by NovaSeq sequencing were debarcoded using vendor software

(bcl2fastq2 v2.20.0) from Illumina. Clonal reads were abandoned, and low sequencing quality tails were trimmed using Phred quality score 30 as the threshold. The cleaned reads from Illumina sequencing were assembled de novo within each barcode group using the Ensemble assembler [11] to merge them into longer contigs. The assembled contigs, as well as singlets, were compared to an in-house viral proteome database using BLASTx with an E-value cutoff of 10^{-5} [12, 13]. The viral proteome database used in BLASTx was compiled using NCBI virus reference proteome (<http://ftp.ncbi.nih.gov/refseq/release/viral/>) to which was added viral proteins sequences from NCBI nr fasta file (based on annotation taxonomy in Virus Kingdom). The database is updated once a year, and the version used in this study was updated in October 2024. Geneious Prime was used with parameters (minimum size: 100; start codon: ATG) to predict potential open reading frames (ORFs). The annotations of these ORFs were assigned based on comparisons with the built-in CDD v3.21 database within the Conserved Domain Database (CDD) [14].

Phylogenetic analysis

To elucidate phylogenetic relationships among hucaurvirus identified in this study and the best matched RNA viruses in GenBank database, related sequences were downloaded from the GenBank database. Protein sequences were aligned using MUSCLE in MEGA-X [15]. Sites containing more than 50% gaps were temporarily removed from the alignments. Maximum likelihood trees were then constructed using IQ-TREE v1.6.12 [16]. All phylogenetic trees were created using IQ-TREE with 1,000 bootstrap replicates (-bb 1000) and the ModelFinder function (-m MFP). Interactive Tree Of Life (iTOL) was used for visualizing and editing phylogenetic trees [16].

Hucaurvirus genome confirmation

To validate the definitive presence of the nucleic acid sequence of hucaurvirus in the original specimens, we conducted independent nucleic acid extractions from 10 samples of the primary library using Trizol reagent, a single-component extraction reagent without supplementary consumables. Briefly, 200 μ L of each sample was thoroughly mixed with 1 mL of Trizol, followed by incubation at room temperature for 5 min to lyse the cells. Next, 0.2 mL of chloroform was added to the mixture, which was then shaken vigorously for 30 s and incubated for another 2 min at room temperature. The mixture was subsequently centrifuged at $12,000\times g$ for 15 min at 4 °C to separate the phases. The upper aqueous phase (containing RNA) was transferred to a new tube, and an equal volume of isopropanol was added to precipitate the RNA at -20 °C for at least 1 h. After re-centrifugation

to form an RNA pellet, the supernatant was discarded. The pellet was washed with 75% ethanol, centrifuged again, and the ethanol was aspirated. Following a brief air-drying step, the RNA was resuspended in RNase-free water. Following nucleic acid isolation, reverse transcription was performed to synthesize complementary DNA strands using SuperScript III reverse transcriptase (Invitrogen) and 100 pmol of a random hexamer primer as described in the library construction. Subsequent PCR screening was implemented using virus-specific primers targeting within the RNA-dependent RNA polymerase (RdRP) gene of the hucaurvirus. The nucleic acids from the original sample pool used for library construction were also included in the PCR screening as a positive control. To confirm the genome sequence generated by sequence reads assembly (Fig. 1B), 7 sets of nested primers were designed based on the genome sequence, here the PCR fragments amplified by adjacent inner sets of primers have a 50–70 bp overlap region (Table 1). PCR amplification was performed using nucleic acids from hucaurvirus-positive samples as templates. These primers information and PCR reaction parameters were shown in Table 1. Phanta Max Master Mix PCR Kit (Vazyme, China) was used in both PCR screening and genomic fragment amplification. Following purification, PCR products were subjected to Sanger sequencing performed by a professional sequencing service provider (Sangon, Shanghai).

Results

The overall viral sequences detected in the CSF samples

The viral metagenomic library constructed from the 10 CSF samples generated a total of 34,130,726 sequence reads through Illumina NovaSeq sequencing platform. Sequence reads and assembled contigs were respectively compared to the GenBank nonredundant (nr) viral protein database with BLASTx (E value of $<10^{-5}$) and 9,233 (0.027%) sequence reads showed significant sequence similarities to viral sequences. Ten groups of virus were detected, with the most abundant being *Retroviridae* (7,445 reads, 80.63%), followed by the *unclassified RNA virus* (1,203 reads, 1.75%), *Siphoviridae* (195 reads, 2.11%), *Myoviridae* (192 reads, 1.48%), *Podoviridae* (64 reads, 0.69%), *Microviridae* (48 reads, 0.52%), *Inoviridae* (31 reads, 0.36%) *Picobirnaviridae* (26 reads, 0.50%), *Picornaviridae* (21 reads, 0.23%), and *unclassified Caudovirales* (8 reads, 0.09%) (Fig. 1A). After sequence assembly, sequence analysis revealed that all the sequence reads matched to *Retroviridae* are endogenous retroviral genes originating from the human genome, which were not included in the subsequent analysis. The remaining viruses that can infect eukaryotes, which may potentially pose a threat to humans, including viruses belonging to *Picornaviridae*, *Picobirnaviridae*, and *unclassified*

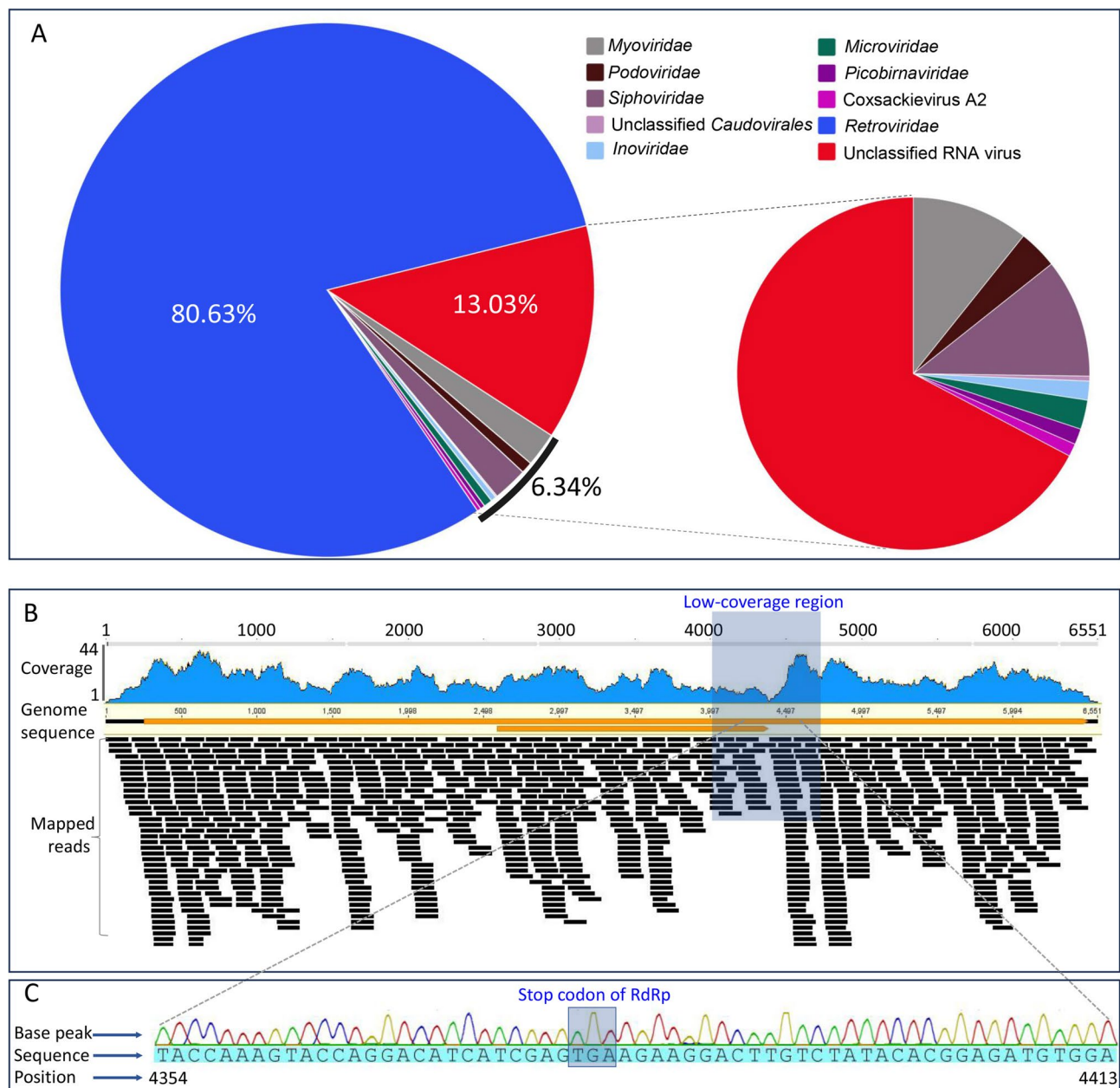


Fig. 1 The composition of the viral community within the CSF sample library and the assembly of the viral genome sequence of hucavirus. **A** The composition of the viral community in the library and the relative abundance of the detected viral reads. **B** Mapping the reads from the raw data back to the genome, where the nearly complete genome sequence of hucavirus is used as a reference. **C** Sanger sequencing chromatogram of low-coverage regions in reads mapping

RNA virus, were included as the key objects of analysis. Intriguingly, these 1,023 sequence reads belonging to *unclassified RNA virus* could be assembled into a contig with a length of 6551 bp (Fig. 1B), where the average depth obtained for the assembled genome is 21.37. Since all sequence reads belonging to unclassified RNA viruses generated by BLASTx can be fully mapped to this contig, and this contig contains complete two ORFs encoding the full-length RdRp protein and viral capsid protein, respectively, we therefore consider this 6551 bp contig to

potentially represent the complete genomic sequence of an unclassified RNA virus. BLASTn analysis of this 6551 bp contig revealed no significantly similar sequences in public databases, suggesting that this sequence may represent the genetic material of a novel virus. Further BLASTx search results indicated that the best matches for this sequence are several unclassified RNA viruses annotated as noda-like viruses and showed the highest sequence identity of 39.2% to the RdRp region of an unclassified RNA virus (*Ixodes ricinus* noda-like-virus 1,

Table 1 Primers' information and reaction parameters for PCR screening and nested PCR for genome amplification

Primer Purpose		Forward Primer (5'–3')	Reverse Primer (5'–3')	Target Fragment Length (bp)	Genome position of the fragment	Annealing Temperature (°C)	Cycle Number
PCR screening		TCCGTGTTGCTTCATCGAT	CCTGCTCCAACCTGAGTCGT	673	2802–3474	52	35
Genome amplification of fragment 1	Outer	GTTGTCCGGAACAGGGACT	AAGTGCTTGCACTGCCATTG	636	77–712	50	28
	Inner	AGGTTGCAGCACCGAAAGTA	TGCTGAGCTGCATGTGTCT	410	222–631	55	35
Genome amplification of fragment 2	Outer	CACGTAACACCCTATGGCGT	CTGCAAGTCCTGACTCTGCA	814	389–1202	50	28
	Inner	GAGGCAGTATGTAACGTGCC	TGCGAGTCTGGTAATGGCTT	616	535–1150	55	35
Genome amplification of fragment 3	Outer	GACCCGAGCCAGGAATTCAA	GACAGTGCTGAAGGCTTCCT	1435	929–2363	50	28
	Inner	TGCTGCTTCTGGAATCACGT	CGTCGCTAGCGCTTGATCT	1070	1081–2150	55	35
Genome amplification of fragment 4	Outer	GATAGCTTGGACGACCTGGG	CACCTGAAGCCATGCTGGAT	1597	2027–3623	50	28
	Inner	GAAGTGACACAGAAAGCGG	TCTGTTGTCTACCCCCAGA	1421	2078–3498	55	35
Genome amplification of fragment 5	Outer	ACCGGGCAAAGATAAGGCAT	GGGTACGTAATGGGCTAGG	1116	3331–4446	50	28
	Inner	CTATAAGGACACTGGAGCGTT	TAGTTCTTCAGGACCGGGG	923	3388–4310	55	35
Genome amplification of fragment 6'	Outer	GCTCCACAGAGTAGGAACGG	AGTGGTGCGTTGTGACTGAT	1254	4193–5446	48	28
	Inner	AAGTGTCGGACGTGGAACAA	GGGGTGTTTGGTGCAACTTC	1137	4243–5379	53	35
Genome amplification of fragment 7'	Outer	AGGGCTCTATTGGGCAACAC	CCTCTTGAGTCTCTCCGGC	1211	5248–6458	48	28
	Inner	GCTGGTGACCATTCAATGC	AACTAGTGACAAACGGGCC	910	5315–6224	53	35

Genbank accession No. USL85435) detected in tick from Croatia [17]. We then named this virus as hucaurvirus (derived from “**H**uman **C**SF-associated **u**nclassified **R**NA virus”).

PCR screening and genome confirmation

To validate the presence of this hucaurvirus within the original CSF samples, we employed TRIzol reagent to individually extract RNA from ten CSF samples in the original sample library. Subsequently, PCR screening was carried out using primers specifically designed according to the genome of hucaurvirus. Ten individual samples and the sample pool of the original library were all included in the PCR screening. Result of the screening indicated that in addition to the original sample pool being positive for this virus, one of the ten samples was also tested positive (Fig. S1A). This hucaurvirus-positive CSF sample was from a 6-year-old male child with encephalitis of unknown etiology, whose clinical symptoms included a body temperature of 39 °C, headache, and projectile vomiting. The bands obtained from the PCR reaction were sent for Sanger sequencing which showed that the PCR products represented the gene sequence of the RdRp region of this hucaurvirus. This finding corroborated the existence of the nucleic acid sequence of hucaurvirus within a CSF sample. PCR amplification products in genome confirmation (Fig. S1B and C) were subjected to Sanger sequencing, and 6 resulting sequences are identical to the original genomic sequences except the first PCR fragment had a nucleic acid site that is inconsistent with the originally assembled genome sequence. However, after we checked the genome sequence assembled from mNGS reads, we found that the sequencing depth at this site exceeds 40X, with high assembly quality.

Therefore, we believe that this mutation may have been introduced during the PCR amplification, and we still consider the originally assembled sequence to be the natural sequence. These results also confirmed the existing of stop codon of the small ORF encoding RDRP (Fig. 1C).

Phylogenetic and sequence analysis

Due to the high sequence divergence of other protein-coding regions in hucaurvirus compared to known viruses and the relatively conserved nature of RdRp, the phylogenetic analysis in this study was performed using the conserved RdRp protein sequences. For a preliminary phylogenetic classification, RDRP protein sequences in the phylogenetic analysis included the representative members from 20 well-defined viral families within the phylum *Kitrinoviricota*, with 2–5 representative sequences selected for each family, several representative members of unclassified *Kitrinoviricota*, and the 4 best matches from BLAST searches in GenBank. Phylogenetic analysis showed that hucaurvirus closely clustered with the four best-matched viruses from BLAST search annotated as noda-like viruses, forming a cluster that is adjacent to the cluster of the *Carmotetraviridae* family (Fig. 2). Although hucaurviruses cluster with several noda-like viruses, they were not closely related to the *Nodaviridae*; instead, they are genetically closely related to the *Carmotetraviridae*, which suggests that these so called “noda-like viruses” are not real nodaviruses. Therefore, to elucidate the genetic relationship among hucaurvirus, noda-like viruses, *Carmotetraviridae*, *Nodaviridae* and *Tombusviridae*. Representative RdRp amino acid sequences of these groups of viruses were used to reconstruct phylogenetic tree, some unclassified RNA virus sequences, despite having relative lower



Fig. 2 Phylogenetic analysis based on the amino acid sequences of RdRp (RNA-dependent RNA polymerase) of hucaurivirus and representative viruses in the phylum *Kitrinoviricota*. The sequence identified in this study was marked with a red dot. The scale bar represents genetic distance

sequence identities (<30%) to hucaurivirus sequences but showing high coverage (>70%) in BLAST searches, were also included in the analysis. Viruses belonging to *Flaviviridae* were used as the outgroup. Results indicated that, being similar to the above overall phylogenetic tree, hucaurivirus clustered closely with the noda-like viruses, forming a separate clade that is genetically closely related to members of the *Carmotetraviridae* but distantly related to those of the *Nodaviridae*. Several unclassified RNA viruses cluster on the periphery of hucaurivirus and *Carmotetraviridae*, forming an independent clade

that may constitute a new distinct viral family within the order *Tolivirales*. (Fig. 3A).

Genome sequence and encoding strategies

The nearly complete genome of hucaurivirus has sequence length of 6551 bp. The genomic sequence contains a large ORF spanning 6240 bp, which encodes a polyprotein of 2079 amino acids. However, upon analysis of this polyprotein, it was found that while the protein includes a virus structural protein-like domain at its C-terminal region, it lacks the RNA-dependent RNA polymerase (RdRp) protein domain. Instead, the RdRp protein is

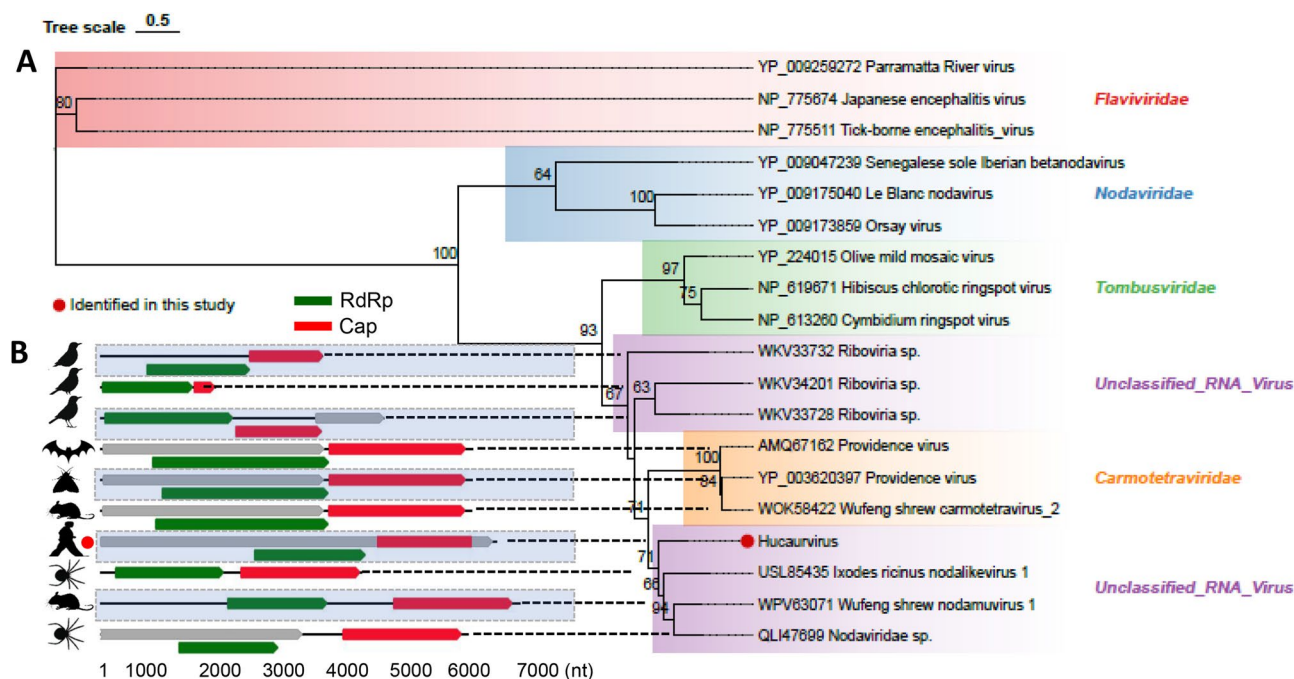


Fig. 3 Phylogenetic analysis, genome structure and detecting resources of hucaurvirus and its and its closely related unclassified RNA viruses. **A** Phylogenetic analysis based on the amino acid sequence of RdRp reveals the genetic relationship between hucaurvirus and its close relatives in GenBank. **B** Genome structure and detecting resources. The sequence identified in this study was marked with a red dot. The green bar represents the RdRp region, and the red bar represents the possible capsid region. The detecting resource information of the viruses closely related to hucaurvirus was also marked on the left side of each genome. The scale bar represents genetic distance

encoded by a smaller ORF of 1979 bp overlapping within the large ORF (Figs. 1B and 3B). This genomic encoding strategy is not only inconsistent with those of *Carmotetraviridae*, but also differs from the genomic organization of all closely related strains on the phylogenetic tree (Fig. 3B). Although this hucaurvirus clusters with these viruses in the phylogenetic tree, they not only exhibit significant differences in genomic structure but also show diversity in their virus detection sources, which are derived from small rodents, arthropods, birds, and bats respectively (Fig. 3B). Additionally, their identities in the RDRP protein sequence to their closely related relatives are very low. Specifically, the RDRP identities between hucaurvirus and viruses belonging to the family *Carmotetraviridae* as well as unclassified RNA viruses ranges only from 21.9 to 39.2%. (Fig. 4).

Discussion

In recent years, the development of NGS technologies has revolutionized the field of virology. NGS allows for the comprehensive analysis of nucleic acids in a sample without prior knowledge of the sequences, leading to the discovery of numerous unclassified RNA viruses in various types of samples [7, 8, 18, 19]. These unclassified RNA viruses exhibit remarkable host diversity. They have been detected in a wide array of hosts, spanning from prokaryotes to eukaryotes [8, 19, 20]. In the eukaryotic

realm, they have been identified in both plants and animals, where some are detected in invertebrates like insects, while others infect vertebrates including mammals and birds [21–23]. This broad host range implies complex ecological interactions. Viruses with such diverse host tropisms may play crucial roles in ecosystem dynamics, potentially influencing the population sizes and behaviors of different host species. The identification of a novel unclassified RNA virus in the CSF of a patient with unexplained meningoencephalitis represents an advance in our understanding of both viral diversity and the etiological spectrum of neuroinflammatory diseases.

To our knowledge, this study is the first to detect an unclassified RNA virus in the cerebrospinal fluid of children with encephalitis of unknown etiology, which expands the range of tissue types from which unclassified RNA viruses have been detected. Unclassified RNA viruses have been detected in a variety of tissue types. In human and animal studies, unclassified RNA viruses have been detected in respiratory tract, blood, and other organs [9, 24, 25]. In environmental samples, they can be associated with organic - rich tissues or materials, which may serve as a reservoir and a source of transmission for these viruses [26]. The ability to detect these viruses in different tissue types is crucial for understanding their routes of infection, replication sites, and potential for systemic spread..

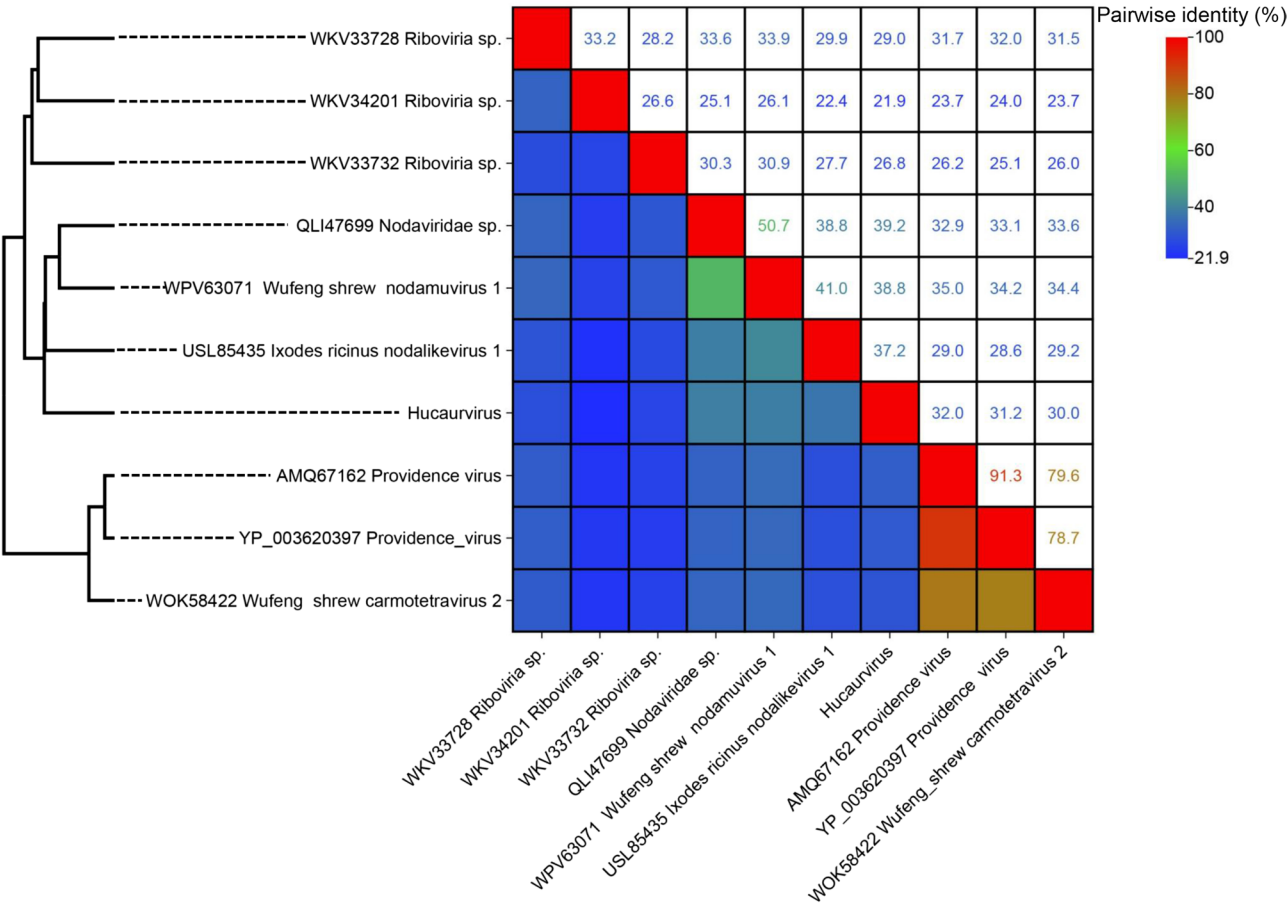


Fig. 4 Sequence similarity between hucaurvirus and its closely closely related relatives based on the amino acid sequence of RdRp. This analysis only included those viruses that are closely related to hucaurvirus, and their virus names and GenBank numbers were marked

In phylogenetic analysis, although the hucaurvirus identified in our study clusters with some unclassified RNA viruses referred to as “noda-like viruses”, their genomic structures differ significantly, and the sequence similarity between them is extremely low. These findings indicate that they do not actually belong to the same viral group. The discrepancy in genomic architecture, combined with the low nucleotide sequence homology, suggests distinct evolutionary trajectories and biological characteristics, supporting the conclusion that they should be classified into separate viral taxa rather than being grouped under the same category. Previous lots of studies also shown that unclassified RNA viruses often occupy unique positions in the viral phylogenetic tree, indicating their distinct evolutionary lineages [6]. Although some unclassified viruses may share certain genetic features with known RNA virus families, suggesting a common ancestry, many also possess unique genetic signatures that distinguish them from recognized virus groups. For example, in the study of RNA viromes from Chinese terrestrial sites, the newly identified unclassified RNA viruses formed distinct clades that were not

affiliated with any known viral families, highlighting their novel evolutionary origins [27]. These genetic and evolutionary relationships are important for tracing the origin and spread of viruses, as well as predicting their potential for host adaptation and emergence..

Although we have temporarily classified hucaurvirus as an unclassified RNA virus, phylogenetic analysis indicates that it may potentially belong to the order *Tollivirales* (Fig. 2), members of which primarily infect plants and insects. Among the several closely related strains of hucaurvirus reflected in phylogenetic analysis, Providence virus has a relatively clear research background, which belongs to *Carmotetraviridae* and is a small, non-enveloped insect RNA virus that was originally identified as a persistent infection in the corn earworm, *Helicoverpa zea* (H. zea), midgut cell line [28]. Recently, Providence virus has been shown to infect and persistently replicate in the human cervical cancer (HeLa) cell line [29], which suggests that the host range of this virus may not be limited to insects. In the present study, the detection of hucaurvirus related to the Providence virus in the

human CSF sample also suggests that the hosts of such viruses may include humans.

Currently, the diagnosis of encephalitis often faces challenges, with up to 50% of cases remaining undiagnosed [2]. Incorporating this new virus into diagnostic algorithms, perhaps through the development of specific PCR-based or serological tests, can improve diagnostic accuracy. In this study, we detected an unclassified RNA virus in clinical samples from patients with encephalitis of unknown etiology, which may indicate that such viruses might have a potential association with encephalitis. Nonetheless, the relationship between unclassified RNA viruses and diseases is still largely under-explored. Some unclassified RNA viruses may be associated with mild or asymptomatic infections, while others could potentially cause severe diseases [30, 31]. However, in other systems, unclassified RNA viruses may be linked to diseases such as respiratory illnesses, gastrointestinal disorders, or even neurodegenerative diseases. The lack of classification and in-depth study of these viruses makes it challenging to accurately assess their disease-causing potential and develop appropriate diagnostic and therapeutic strategies.

Conclusion

This study identifies a novel unclassified RNA virus, hucaurvirus, in CSF from pediatric meningoencephalitis patients via mNGS. Its 6.5 kb positive-sense RNA genome features a unique architecture: a large ORF encoding a capsid polyprotein and an overlapping small ORF for RdRp, distinct from related noda-like viruses and *Carmotetraviridae*. Phylogenetic analysis shows clustering with noda-like viruses but low homology (< 39.2%), supporting it as a novel taxon. PCR confirms its presence in one of ten CSF samples, expanding unclassified RNA viruses' known tissue tropism. Future work should focus on culture system establishment, large-scale epidemiological studies, reservoir exploration, and taxonomic refinement to clarify its role in CNS infections.

Abbreviations

CSF	cerebrospinal fluid
CNS	central nervous system
NGS	Next Generation Sequencing

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1186/s12866-025-04360-2>.

Supplementary Material 1.

Supplementary Material 2.

Supplementary Material 3.

Authors' contributions

All authors participated in the design, interpretation of the studies and analysis of the data and review of the manuscript; WZ, CZ, and YL contributed to the conception and design; JZ, YZ, WL, YC, SY and JX contributed to the collection and assembly of data; JZ, PW, SY, CZ and WZ contributed to the data analysis and interpretation.

Funding

This research was financially supported by National Natural Science Foundation of China (No. 82341106), Medical Education Collaborative Innovation Fund of Jiangsu University (No. JDY2022008), and Jiangsu Provincial Medical Key Discipline Cultivation Unit (No. JSDW202239).

Data availability

The NGS raw data supporting this study's findings have been deposited in the NCBI Sequence Read Archive (SRA) under the BioProject accession number PRJNA1229836, BioSample accession number SAMN47143759, and SRA accession number SRX27856860. The genome sequence reported in this study is publicly available in the GenBank database under the accession numbers PV252795. These datasets are openly accessible through the NCBI database.

Declarations

Ethics approval and consent to participate

This study adhered to the Declaration of Helsinki and obtained approval from the Ethics Committee of Jiangsu University (Approval No. JSDX1007002). Informed consent to participate was obtained from the parents or legal guardians.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

Received: 1 April 2025 / Accepted: 26 August 2025

Published online: 25 November 2025

References

1. Granerod J, Ambrose HE, Davies NWS, Clewley JP, Walsh AL, Morgan D, et al. Causes of encephalitis and differences in their clinical presentations in England: A multicentre, population-based prospective study. *Lancet Infect Dis*. 2010;10:835–44.
2. Glaser CA, Honarmand S, Anderson LJ, Schnurr DP, Forghani B, Cossen CK, et al. Beyond viruses: clinical profiles and etiologies associated with encephalitis. *Clin Infect Dis*. 2006;43:1565–77.
3. Wilson MR, Sample HA, Zorn KC, Arevalo S, Yu G, Neuhaus J, et al. Clinical metagenomic sequencing for diagnosis of meningitis and encephalitis. *N Engl J Med*. 2019;380:2327–40.
4. Miller S, Naccache SN, Samayoa E, Messacar K, Arevalo S, Federman S, et al. Laboratory validation of a clinical metagenomic sequencing assay for pathogen detection in cerebrospinal fluid. *Genome Res*. 2019;29:831–42.
5. Wolf YI, Kazlauskas D, Iranzo J, Lucia-Sanz A, Kuhn JH, Krupovic M, et al. Origins and evolution of the global RNA Virome. *MBio*. 2018. 9(6):e02329-18. <https://doi.org/10.1128/mBio.02329-18>.
6. Shi M, Lin X-D, Tian J-H, Chen L-J, Chen X, Li C-X, et al. Redefining the invertebrate RNA virosphere. *Nature*. 2016;540:539–43.
7. Dong X, Meng F, Zhou C, Li J, Hu T, Wang Y, et al. Enormous diversity of RNA viruses in economic crustaceans. *mSystems*. 2024; 9(10):e0101624.
8. Nakagawa S, Sakaguchi S, Ogura A, Mineta K, Endo T, Suzuki Y, et al. Current trends in RNA virus detection through metatranscriptome sequencing data. *FEBS Open Bio*. 2023;13:992–1000.
9. Zhang W, Yang S, Shan T, Hou R, Liu Z, Li W, et al. Virome comparisons in wild-diseased and healthy captive giant pandas. *Microbiome*. 2017;5:90.
10. Zhao M, Yue C, Yang Z, Li Y, Zhang D, Zhang J, et al. Viral metagenomics unveiled extensive communications of viruses within giant pandas and their associated organisms in the same ecosystem. *Sci Total Environ*. 2022;820:153317.

11. Deng X, Naccache SN, Ng T, Federman S, Li L, Chiu CY, et al. An ensemble strategy that significantly improves de novo assembly of microbial genomes from metagenomic next-generation sequencing data. *Nucleic Acids Res.* 2015;43:e46–46.
12. Bao S, Wang H, Li W, Wu H, Lu C, Yong L et al. Viral metagenomics of the gut Virome of diarrheal children with rotavirus A infection. *Gut Microbes.* 2023;15(1):2234653.
13. Lu X, Gong G, Zhang Q, Yang S, Wu H, Zhao M, et al. Metagenomic analysis reveals high diversity of gut Viromes in Yaks (*Bos grunniens*) from the Qinghai-Tibet plateau. *Commun Biol Springer US.* 2024;7:1–12.
14. Wang J, Chitsaz F, Derbyshire MK, Gonzales NR, Gwadz M, Lu S, et al. The conserved domain database in 2023. Oxford University Press. *Nucleic Acids Res.* 2023;51:D384–8.
15. Kumar S, Stecher G, Li M, Knyaz C, Tamura K. MEGA X: molecular evolutionary genetics analysis across computing platforms. *Mol Biol Evol.* 2018;35:1547–9.
16. Minh BQ, Schmidt HA, Chernomor O, Schrempf D, Woodhams MD, Von Haeseler A, et al. Iq-tree 2: new models and efficient methods for phylogenetic inference in the genomic era. *Mol Biol Evol.* 2020;37:1530–4.
17. Sameroff S, Tokarz R, Vucelja M, Jain K, Oleynik A, Boljefi M et al. Virome of *Ixodes ricinus*, *Dermacentor reticulatus*, and *Haemaphysalis concinna* Ticks from Croatia. 2022 14(5):929.
18. Lu X, Ji L, Wang H, Zhang Q, Wang X, Liu Y, et al. Highly diverse RNA viruses and phage sequences concealed within birds. *Microbiol Spectr Am Soc Microbiol.* 2024;12:1–19.
19. Zhang YZ, Chen YM, Wang W, Qin XC, Holmes EC. Expanding the RNA virosphere by unbiased metagenomics. *Annu Rev Virol.* 2019;6:119–39.
20. Ramos EDSF, Ribeiro G, de O, Villanova F, Milagres FA, de Brustulin P, Araújo R. Composition of eukaryotic viruses and bacteriophages in individuals with acute gastroenteritis. *Viruses.* 2021;13:1–12.
21. Yang Z, Wang H, Yang S, Wang X, Shen Q, Ji L, et al. Virome diversity of ticks feeding on domestic mammals in China. *Virol Sin Authors.* 2023;38:208–21.
22. Shan T, Yang S, Wang H, Wang H, Zhang J, Gong G, et al. Virome in the cloaca of wild and breeding birds revealed a diversity of significant viruses. *Microbiome.* 2022;10:1–21.
23. Yang S, Mao Q, Wang Y, He J, Yang J, Chen X, et al. Expanding known viral diversity in plants: virome of 161 species alongside an ancient canal. *Environ Microbiome.* 2022;17:1–15.
24. Zhuang Z, Qian L, Lu J, Zhang X, Mahmood A, Cui L, et al. Comparison of viral communities in the blood, feces and various tissues of wild brown rats (*Rattus norvegicus*). *Heliyon.* 2023;9:e17222 Elsevier Ltd.
25. Obraitis D, Li D. Blood Virome research in myalgic encephalomyelitis/chronic fatigue syndrome: challenges and opportunities. *Curr Opin Virol.* 2024;68–69:101437.
26. Lu J, Yang S, Zhang X, Tang X, Zhang J, Wang X, et al. Metagenomic analysis of viral community in the Yangtze River expands known eukaryotic and prokaryotic virus diversity in freshwater. *Virol Sin.* 2022;37:60–9.
27. Chen YM, Sadiq S, Tian JH, Chen X, Lin XD, Shen JJ, et al. RNA viromes from terrestrial sites across China expand environmental viral diversity. *Nat Microbiol.* 2022. <https://doi.org/10.1038/s41564-022-01180-2>.
28. Pringle FM, Johnson KN, Goodman CL, McIntosh AH, Ball LA. Providence virus: a new member of the tetraviridae that infects cultured insect cells. *Virology.* 2003;306:359–70.
29. Jiwaji M, Short JR, Dorrington RA. Expanding the host range of small insect RNA viruses: Providence virus (Carmotetraviridae) infects and replicates in a human tissue culture cell line. *J Gen Virol.* 2016. 97(10):pp. 2763–8.
30. Cordey S, Laubscher F, Hartley MA, Junier T, Pérez-Rodríguez FJ, Keitel K, et al. Detection of dicistroviruses RNA in blood of febrile Tanzanian children. *Emerg Microbes Infect.* 2019. <https://doi.org/10.1080/22221751.2019.1603791>.
31. Siqueira JD, Dominguez-Bello MG, Contreras M, Lander O, Caballero-Arias H, Xutao D, et al. Complex virome in feces from amerindian children in isolated Amazonian villages. *Nat Commun.* 2018;9:4270.

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